

Course Code	Course Title	L	T	P	C
BCSE407L	Computer Vision	3	0	0	3
Pre-requisite	NIL	Syllabus version			
		1.0			
Course Objectives					
1. To solve real world problems with image or video as input.					
2. To make use of low level image processing algorithms to provide information about the scene.					
3. To emphasize on computer vision applications					
Course Outcomes					
At the end of the course the student will be able to					
1. Analyze image formation using digital camera and its principles					
2. Evaluate feature extraction and feature estimation for image or video					
3. Apply 3D vision techniques					
4. Identify the computer vision applications					
Module:1 Low Level Vision 6 hours					
Image Formation – Sampling and Aliasing – Linear Filters and Convolution – Correlation and Patterns – Image Pyramid					
Module:2 Feature Detection and Matching 5 hours					
Points and patches-Feature detectors, Feature descriptors, Feature matching, Feature tracking; Edges: Edge detection and linking; Vanishing points					
Module:3 Segmentation 7 hours					
Active Contours – Split and Merge – Mean Shift and Mode Shift – Normalized cut – Graph cut and Energy based methods – Deep Learning based Segmentation Models for Computer Vision					
Module:4 Motion Estimation and Recognition 7 hours					
Translational alignment – Parametric Motion - Spline-based motion – Optical Flow – Layered Motion- Object Detection – Face Recognition – Scene Understanding					
Module:5 Stereo Correspondence and 3D Reconstruction 6 hours					
Epipolar Geometry – Local Methods – Multi view stereo – Shape from X – Active Range finding – Model based reconstruction					
Module:6 Image Stitching and Image Rendering 6 hours					
Stitching: Motion models, Global alignment, Compositing; Rendering: Layered depth images, Light fields and Lumi graphs, Environment Mattes					
Module:7 Computer Vision Applications 6 hours					
Contour tracking and rotoscoping – Medical Image Segmentation – Video summarization and compression – Stereo based head tracking – Z-keying and background replacement					
Module:8 Recent Trends 2 hours					
Guest lectures from Industry and, Research and Development Organizations					
		Total Lecture hours:		45 hours	
Text Book(s)					
1.	Richard Szeliski, Computer Vision: Algorithms and Applications, Springer-Verlag London Limited, 2011				
Reference Books					